

Gen AI Exchange Hackathon

Team Name : Turbo C++

Team Leader Name : Jaivik Chauhan

Problem Statement : [Student] Generative AI for Demystifying Legal Documents

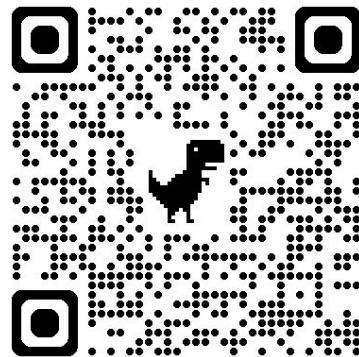
Brief about the prototype:

The TautologyAI prototype demonstrates a streamlined process for legal document analysis. Users can securely upload documents and receive immediate, AI-generated summaries and risk assessments.

The platform's intuitive dashboard enables efficient document management, review, and result export, ensuring accessibility for both legal professionals and general users.

Key Features:

- Rapid document upload and automated processing
- Clear, concise summaries in plain English
- Identification of key risks and clauses
- Experience the prototype to explore advanced legal insights delivered with speed and accuracy.



[Scan or click me
to see working
prototype](#)

[Click me to visit
Git Repository of
the project](#)

How different is it from any of the other existing solutions?

- TautologyAI stands out by combining advanced document processing (Google Document AI) with real-time summarization and risk analysis using state-of-the-art AI models (OpenAI, Gemini).
- Unlike traditional legal tech tools, it offers instant, plain-English summaries and risk identification for uploaded documents, with seamless integration into modern cloud storage (Supabase, Firebase). The user experience is streamlined, secure, and designed for both legal professionals and non-experts.

How will it be able to solve the problem?

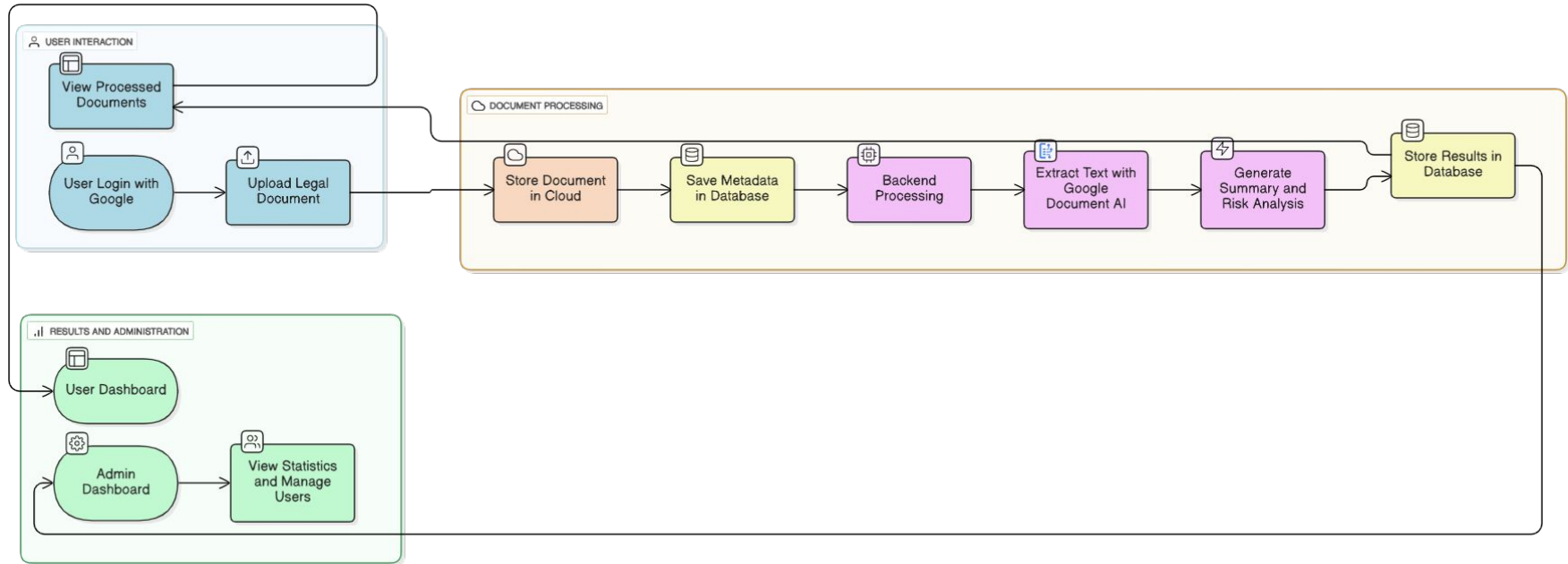
- TautologyAI automates the extraction, analysis, and summarization of complex legal documents. Users simply upload a file, and the system processes it end-to-end: extracting text, analyzing key clauses, and generating actionable summaries and risk highlights. This reduces manual review time, minimizes errors, and makes legal insights accessible to everyone, regardless of legal expertise.

USP of the proposed solution

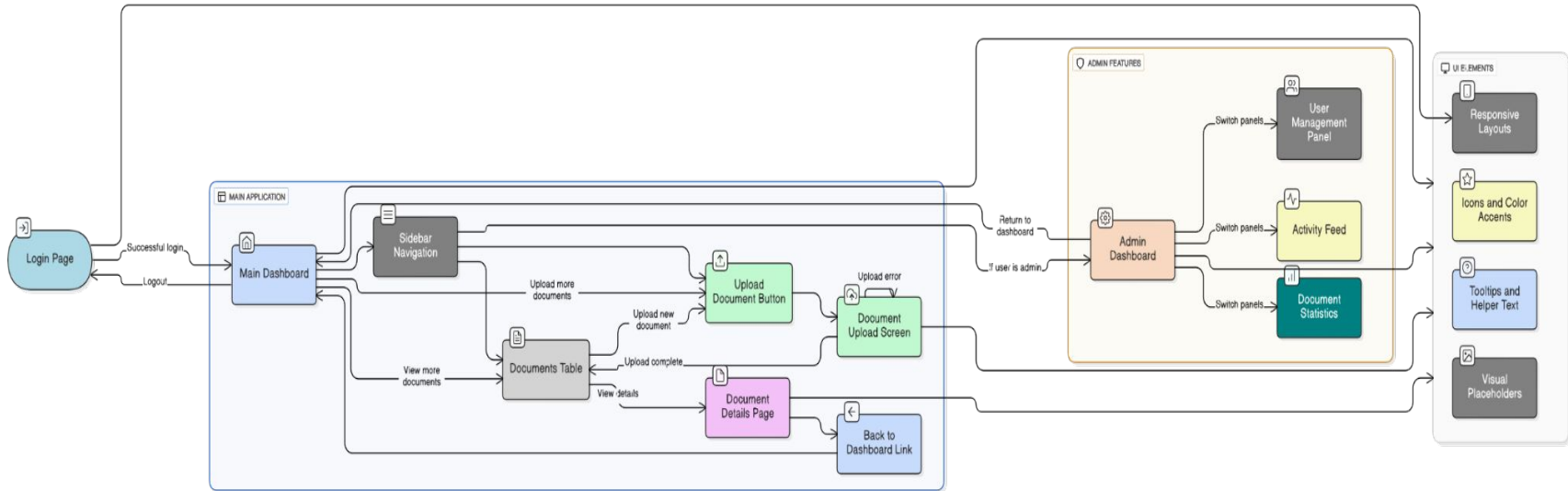
- End-to-end automation: From upload to summary, no manual intervention required.
- AI-powered risk detection: Highlights risky clauses and key parties using the latest LLMs.
- User-centric design: Fast, secure, and easy to use, with instant feedback and cloud integration.
- Scalable architecture: Built for modern web and cloud environments, supporting large volumes and multiple users.

- ❑ **Automated Document Upload:** Users can securely upload legal documents (PDF, DOC, DOCX) via a simple web interface.
- ❑ **AI-Powered Text Extraction:** Advanced models extract text and key information from complex documents.
- ❑ **Instant Summarization:** Generates plain-English summaries and executive bullet points for each document.
- ❑ **Risk Analysis:** Identifies and highlights risky clauses, key parties, and potential red flags.
- ❑ **User Authentication & Activity Tracking:** Secure login via Google, with user activity tracked for insights.
- ❑ **Cloud Storage Integration:** Documents and metadata are stored and managed using Supabase and Firebase.
- ❑ **Admin Dashboard:** Provides statistics, user management, and document oversight for administrators.
- ❑ **Download & Copy Results:** Users can copy or download summaries and extracted text for further use.
- ❑ **Responsive UI:** Modern, mobile-friendly design for seamless user experience.

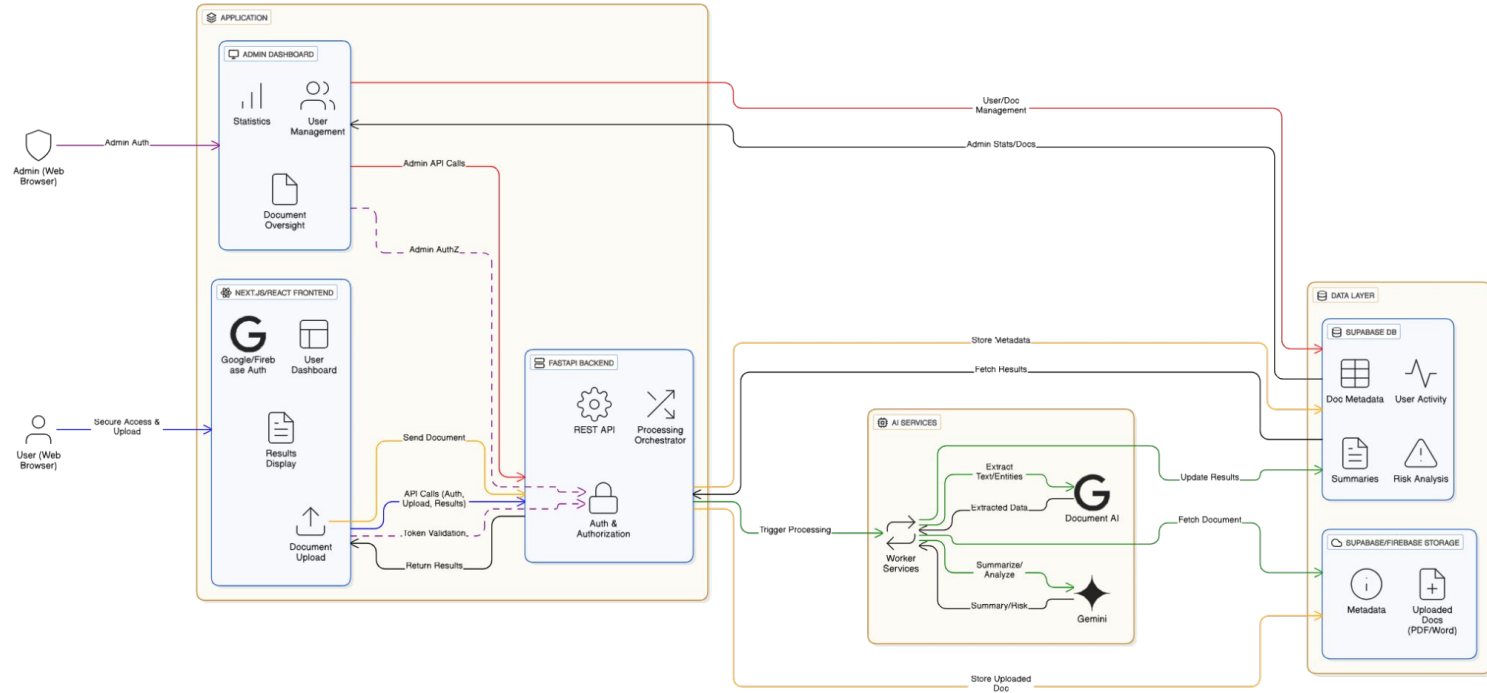
Process flow diagram :



Clear wireframes showcase how users move from uploading documents to viewing instant summaries and risk analysis :



Architecture diagram of the proposed solution



Technologies to be used in the solution :

- **Frontend:** Next.js, React, Tailwind CSS, Radix UI, Shadcn UI, Lucide Icons
- **Authentication:** Firebase Authentication (Google login)
- **Backend:** FastAPI (Python)
- **Cloud Storage & Database:** Supabase (storage and database), Firebase (storage)
- **AI Services:** Google Document AI (text extraction), Gemini (summarization and risk analysis)
- **Other:** Python libraries (pdfplumber, supabase-py, python-dotenv), REST APIs, Uvicorn (server)
- **Dev Tools:** TypeScript, ESLint, PostCSS, Docker (for deployment)
- **Monitoring & Analytics:** Integrated logging, error tracking, and usage analytics for performance and reliability.
- **Testing & Quality Assurance:** Automated testing frameworks and code quality tools to ensure robust, maintainable development.

Estimated Implementation Cost :

- **Cloud services** (Google Document AI, Vertex/Gemini): \$20-\$50/month (depending on usage)
- **Development** (frontend, backend, AI integration): Free Tier (Scaling up project further might apply costs)
- **Domain/Hosting**: Free Tier (Scaling up might cost)

Unique Selling Proposition :

- TautologyAI delivers instant, plain-English summaries and risk analysis for legal documents, powered by advanced AI models. The platform's seamless workflow and secure cloud integration set it apart, making legal insights accessible to everyone.

Future Goals and Scale-Ups :

- Expand support for more document types and languages.
- Integrate advanced AI models for deeper legal analysis and clause extraction.
- Add collaboration features for teams and law firms.
- Enhance security and compliance for enterprise clients.
- Scale infrastructure to handle thousands of users and documents.
- Launch mobile apps for on-the-go access.
- Develop analytics dashboards for user insights and legal trends.
- Integrate with external legal databases and APIs for richer context and references.
- Offer customizable AI models tailored to specific industries or legal domains.
- Implement real-time notifications and workflow automation for document status updates.
- Build partnerships with law firms and enterprise clients for broader adoption.
- Enable multi-language support and cross-border legal document analysis.

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Thank you